

Project Specification: QuillInsight (AI-Assisted Development)

1. Introduction & Project Goal

This project involves building **QuillInsight**, an AI-powered note-taking web application. The application allows users to write, organize, and manage notes while leveraging AI to automatically summarize content, generate tags, and highlight key insights.

AI will be actively integrated throughout the project lifecycle—from planning and scaffolding to testing, debugging, and documentation—demonstrating how AI can accelerate and enhance modern development workflows.

2. Core Application Features

The completed application will include the following features:

Notes Management (for logged-in users)

- Ability to **create, edit, and delete notes**.
- Option to organize notes with **folders and tags**.
- Dashboard view listing all notes with search and filter functionality.

AI-Powered Features

- **Summarization:** Generate concise summaries of long notes.
- **Auto-tagging:** Suggest relevant tags for quick organization.
- **Highlight extraction:** Identify key points or action items from notes.

User Authentication

- User registration and login via **Supabase Auth**.
- Secure access to personal notes only.

Results & Insights Display

- View AI-generated summaries and tags alongside the original note.
- Toggle between raw note and AI-enhanced insights.

3. User Roles within the Application

1. Registered User

- Logs in to the app.
- Creates, edits, deletes, and organizes notes.
- Uses AI features (summarize, tag, extract insights).

2. Anonymous Visitor

- Can test limited features (e.g., demo note summarization).
- Must register to save and persist notes.

4. Technical Foundation

- **Platform:** Web Application
- **Architecture:** Full-stack application built with **Next.js (App Router)**
- **Data Persistence:** Supabase (Postgres, Auth, Storage)
- **Key Functionality:** AI integration for summarization, tagging, and highlights
- **Deployment:** Vercel for frontend + Supabase cloud services

5. AI Tool Integration During Development

AI tools will be incorporated in multiple stages of the project:

Planning & Design

- Use AI chat tools to structure **data models** (e.g., notes schema, tags schema).
- Brainstorm **user flows** and **UI wireframes**.

- Outline Next.js API routes and Supabase interactions.

UI Generation

- Employ AI-powered UI generators (e.g., v0.dev, shadcn/ui) to scaffold reusable UI components such as the note editor, search bar, and dashboard cards.

Code Generation & Assistance

- Leverage AI-native IDE assistants (e.g., Cursor, Copilot) to:
 - Scaffold React components.
 - Write Supabase queries for CRUD operations.
 - Generate TypeScript utility functions for AI API calls.

Testing

- Use AI to propose **unit and integration test cases**.
- Generate Jest/React Testing Library tests for the note editor and Supabase APIs.
- Prompt AI for **edge case discovery** (e.g., empty notes, long text).

Debugging

- Use AI to explain stack traces, propose fixes, and refactor complex functions.
- Prompt AI to optimize API calls and improve performance.

Documentation

- Generate inline comments and **docstrings** for functions and components.
- Keep **README.md** and API documentation up to date using AI-generated summaries from recent commits/diffs.
- Use AI to craft **commit messages** and **PR descriptions**.

6. Chosen Technology Stack

- **Framework:** Next.js (with App Router)
- **Frontend Library:** React + TypeScript
- **Database & Auth:** Supabase (Postgres, Supabase Auth, Storage)
- **Styling:** Tailwind CSS + shadcn/ui components
- **AI Integration:** OpenAI API for summarization, tagging, and insights
- **Deployment Platform:** Vercel (frontend) + Supabase cloud services

7. Project Outcomes

Upon completion of the project, expected deliverables include:

- **Completed Source Code:**

A GitHub repository containing the entire source code of QuillInsight, with commit history reflecting AI-assisted development.

- **Live Deployed Application:**

A publicly accessible web app hosted on Vercel, connected to Supabase.

- **Comprehensive README.md:**

Detailed documentation that explains:

- The project's purpose and features.
- Steps to set up and run locally.
- The technology stack used.
- How AI tools were integrated into the development lifecycle.

- **Demo Presentation:**

A 3–5 minute walkthrough showcasing the app's features and highlighting AI-assisted workflows during development.

QuillInsight Color Palette

Brand Color (Primary)

- **Electric Blue** #3B82F6 (main identity, buttons, highlights)

Supporting Colors

- **Dark Navy** #0F172A (headings, emphasis, dashboard backgrounds)
 - **Soft Gray** #F8FAFC (neutral background for light mode)
 - **Charcoal Gray** #1E293B (neutral background for dark mode)
 - **Accent Purple** #8B5CF6 (AI features, tooltips, secondary actions)
 - **Success Green** #22C55E (status indicators, confirmation)
 - **Warning Amber** #F59E0B (alerts, notifications)
 - **Error Red** #EF4444 (validation errors)
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Themes Setup

Public Pages (Light & Dark Mode)

- Light mode:

- Background → #F8FAFC
 - Text → #0F172A
 - Primary CTA → #3B82F6
 - Secondary CTA → #8B5CF6
- Dark mode:
 - Background → #0F172A
 - Text → #F8FAFC
 - Primary CTA → #3B82F6
 - Secondary CTA → #8B5CF6

Dashboard Themes (5 Variants)

You can let users toggle these in **Settings** → **Personalization**:

1. **Classic Light**

- Background: #F8FAFC
- Text: #0F172A
- Primary: #3B82F6

2. Professional Dark

- Background: #0F172A
- Text: #F8FAFC
- Primary: #3B82F6

3. Modern Purple

- Background: #1E1B4B
- Text: #EDE9FE
- Primary: #8B5CF6

4. Eco Green

- Background: #ECFDF5
- Text: #064E3B
- Primary: #22C55E

5. Minimal Gray

- Background: #F1F5F9
- Text: #1E293B
- Primary: #3B82F6